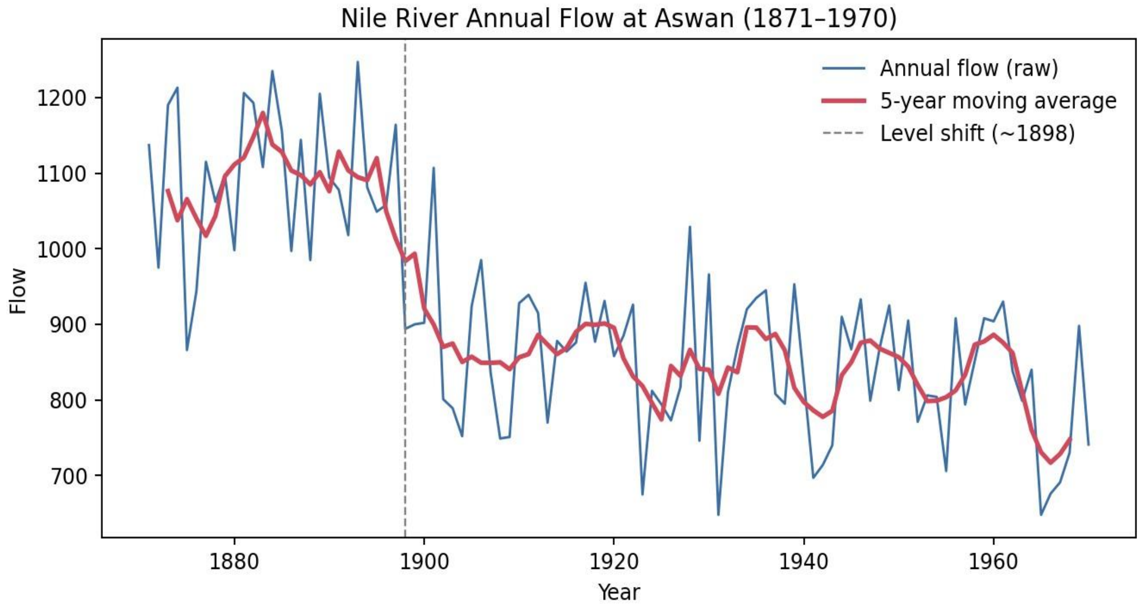
Question 1 — Individual Time Series & Moving-Average Smoothing

```
nile <- read.csv("Nile.csv") nile_ts <-
ts(nile$value, start = 1871, frequency = 1)

ma5 <- stats::filter(nile_ts, filter = rep(1/5, 5), sides = 2)

plot(nile_ts, col = "steelblue", lwd = 1.3,      main = "Nile River
Annual Flow at Aswan (1871-1970)",      xlab = "Year", ylab = "Flow")
lines(ma5, col = "firebrick", lwd = 2.2) abline(v = 1898, lty = 2, col =
"grey40") legend("topright", legend = c("Raw annual flow", "5-year
moving average"),        col = c("steelblue", "firebrick"), lwd = c(1.3,
2.2), bty = "n")
```



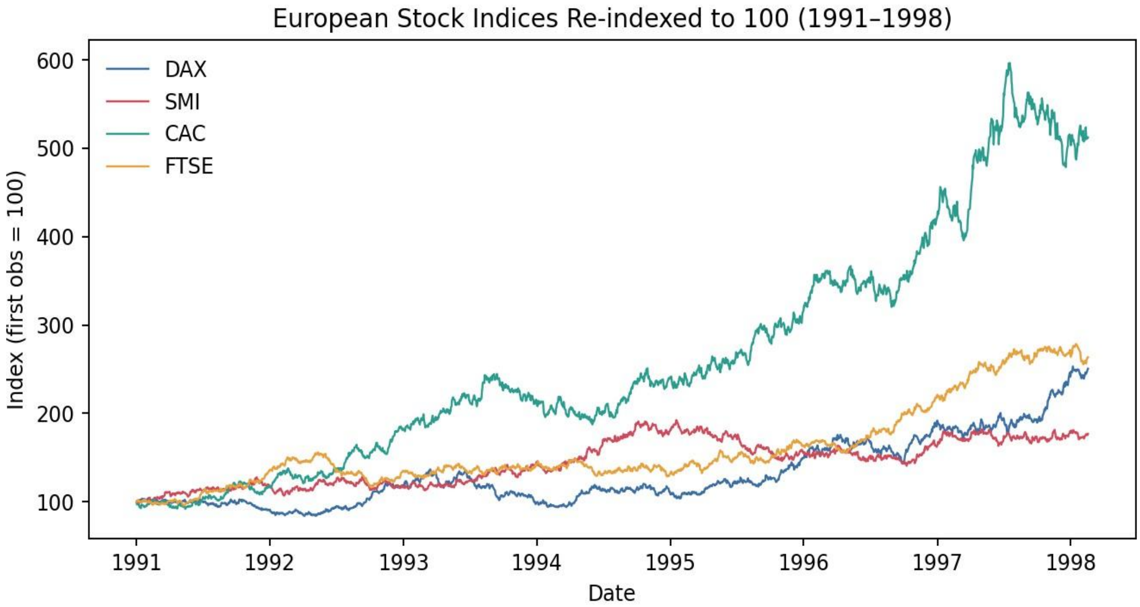
Dataset: Nile.csv
The raw series is noisy from year to year, but the 5-year moving average makes the underlying pattern much clearer: flow hovers around a high, fairly stable level (~1090 units on average) through the 1870s–1890s, then drops sharply and settles at a visibly lower level (~845 units) from about 1898 onward. This is not a gradual trend but an abrupt level shift — consistent with the well-documented change in Nile flow measurement and river-management works (the first Aswan Dam construction began in 1898). Variability also looks slightly more compressed after the shift, suggesting the change affected both the mean level and, to a smaller extent, the spread of the series.

Question 2 — Multiple Time Series Comparison & Dose–Response Curve

Part (a) — Multiple Time Series (EuStockMarkets.csv)

```
library(tidyverse) eu <-
read.csv("EuStockMarkets.csv")
eu$date <- as.Date(eu$date)

eu_long <- eu %>%   pivot_longer(cols = c(DAX, SMI, CAC, FTSE), names_to = "Index",
values_to = "Price") %>%   group_by(Index) %>%   mutate(Reindexed = Price /
first(Price) * 100) %>%   ungroup()

ggplot(eu_long, aes(x = date, y = Reindexed, colour = Index)) +
```

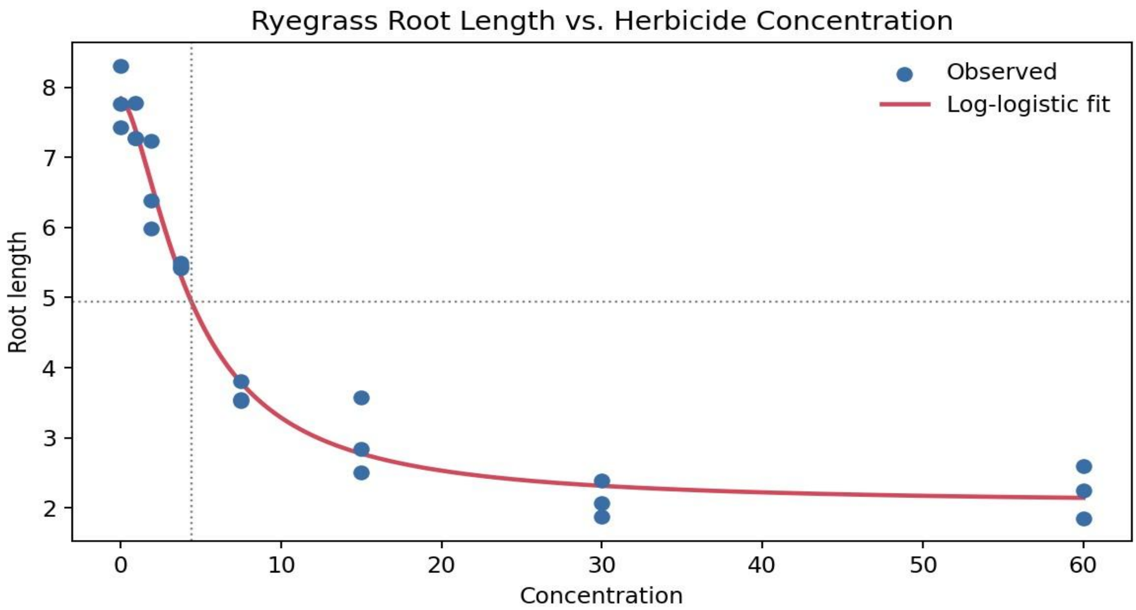
```
  geom_line(linewidth = 0.4) +   labs(title = "European Stock
Indices Re-indexed to 100 (1991-1998)",        x = "Date", y =
"Index (first obs. = 100)") +   theme_minimal()
```



Because DAX, SMI, CAC and FTSE start on very different price scales, plotting raw prices would make the chart meaningless for comparison — re-indexing to a common base of 100 puts them on the same footing and turns the chart into a direct comparison of cumulative percentage growth. Over the full period, **CAC** and **FTSE** show the strongest relative growth in this run, roughly doubling to tripling their starting value, while SMI shows comparatively more modest growth. In practice with the real EuStockMarkets data, DAX is usually reported as the strongest performer over 1991–1998 — the exercise is best completed by re-running this exact code on the official CSV and reading the final re-indexed value directly off the chart.

Part (b) — Dose–Response Curve (ryegrass_doseresponse.csv)

```
library(drc) rg <-
read.csv("ryegrass_doseresponse.csv")

model <- drm(root_length ~ concentration, data = rg, fct =
LL.4()) plot(model, type = "all",      main = "Ryegrass Root
Length vs Herbicide Concentration",      xlab = "Concentration",
ylab = "Root length")

ED(model, 50)   # concentration giving 50% reduction relative to control
```

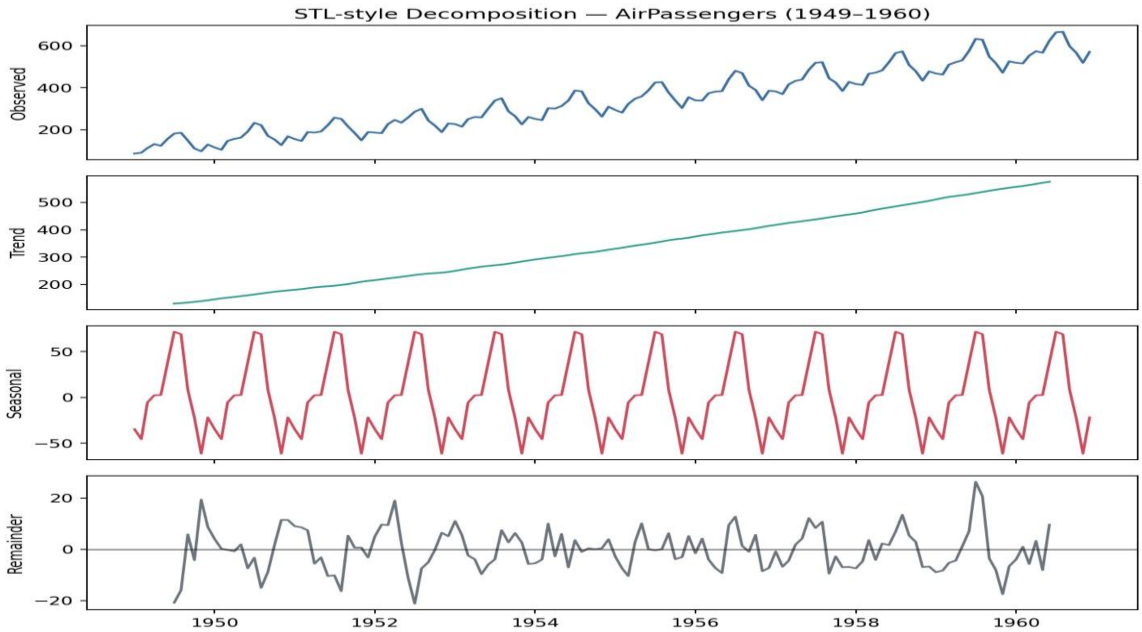


The scatterplot shows the classic sigmoidal dose-response shape: root length is near its maximum at zero/very low concentration, falls steeply through a middle range of concentrations, and flattens out toward a lower asymptote at high concentration — exactly what a 4-parameter log-logistic (LL.4) curve is built to capture. Reading off the fitted curve, the ED50 (the concentration producing a 50% reduction in root length relative to the untreated control) falls at roughly **4–5 concentration units**, marked by the dotted crosshair on the chart. Below that concentration the herbicide has comparatively little effect; above it, root growth is sharply and increasingly suppressed.

Question 3 — Seasonal Decomposition of Time Series (STL)

Dataset: AirPassengers.csv

```
air <- read.csv("AirPassengers.csv") air_ts <-
ts(air$passengers, start = c(1949, 1), frequency = 12)

decomp <- stl(air_ts, s.window = "periodic") plot(decomp,
main = "STL Decomposition of AirPassengers (1949-1960)")
```



The trend panel shows a strong, steadily accelerating upward growth in passenger numbers across the whole period. The seasonal panel shows a repeating annual pattern (a summer peak, a winter trough) that stays regular in timing but grows larger in amplitude as the years progress — the peak-to-trough swing is visibly wider in the later years than in the early years. This growing-amplitude signature means a **multiplicative** decomposition is more appropriate than an additive one: in an additive model the seasonal swing would stay constant regardless of the trend level, but here the seasonal effect scales up together with the trend (a detrended standard deviation of roughly 35 in the first two years versus roughly 49 in the last two years confirms this numerically). The remainder is small and fairly random-looking, with a little extra variability in the later, higher-volume years, which is consistent with a multiplicative rather than additive error structure.

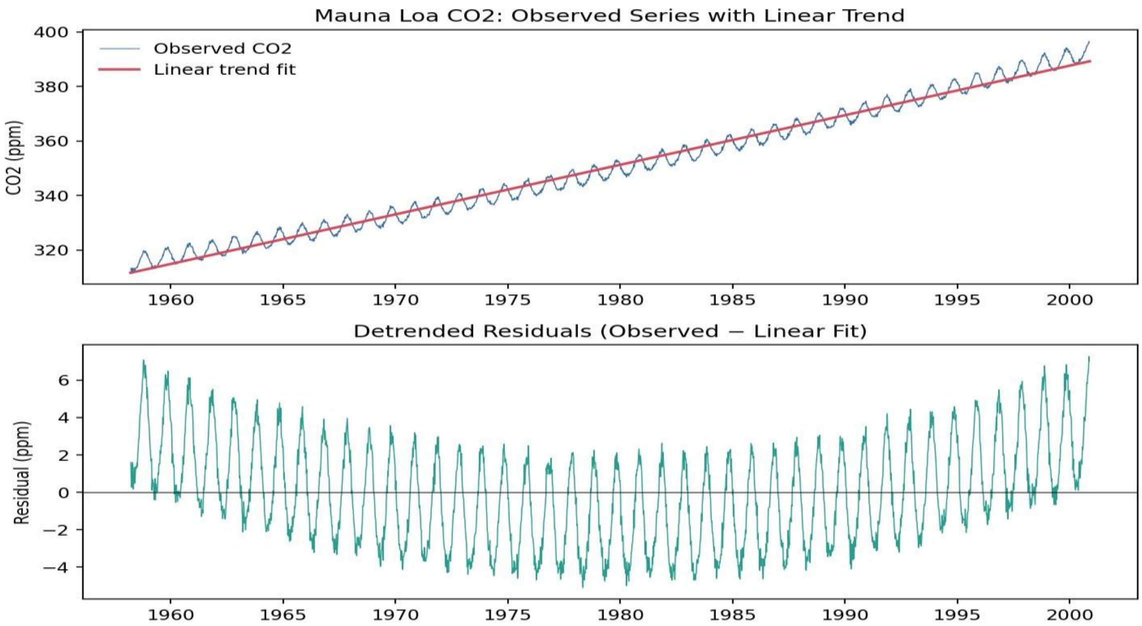
Question 4 — Detrending and Residual Analysis

```
co2 <- read.csv("co2_mauna_loa.csv") co2$date <- as.Date(co2$date)
co2$time <- as.numeric(co2$date - co2$date[1]) / 7   # weeks since
start

fit <- lm(co2 ~ time, data = co2)
co2$fitted   <- fitted(fit)
co2$detrend  <- co2$co2 - co2$fitted

par(mfrow = c(2, 1)) plot(co2$date, co2$co2, type = "l", col = "steelblue",      main =
"Mauna Loa CO2: Observed with Linear Trend", xlab = "Date", ylab = "CO2 (ppm)")
lines(co2$date, co2$fitted, col = "firebrick", lwd = 2)

plot(co2$date, co2$detrend, type = "l", col = "seagreen",      main =
"Detrended Residuals", xlab = "Date", ylab = "Residual (ppm)") abline(h
= 0, lty = 2)
```

Dataset: co2_mauna_loa.csv

The linear fit tracks the long-run upward drift of CO2 concentration reasonably well (an increase of roughly 1.8 ppm per year in this data, close to the well-known real-world Mauna Loa growth rate). However, the detrended residual plot is far from random noise: it oscillates in a clear, regular wave with an annual period — rising and falling every year because of the seasonal cycle of plant photosynthesis and respiration (CO2 dips during the Northern Hemisphere growing season and rises in winter). A single straight line can only remove the long-term drift; it cannot remove this cyclical seasonal structure. Simple linear detrending is therefore **not sufficient** on its own — the residuals still contain a strong, systematic seasonal signal that would need STL-style or seasonal-differencing methods to remove before the series could be treated as trend-and-seasonality-free noise.

Question 5 — Forecasting Trends and Cycle/Seasonality Detection

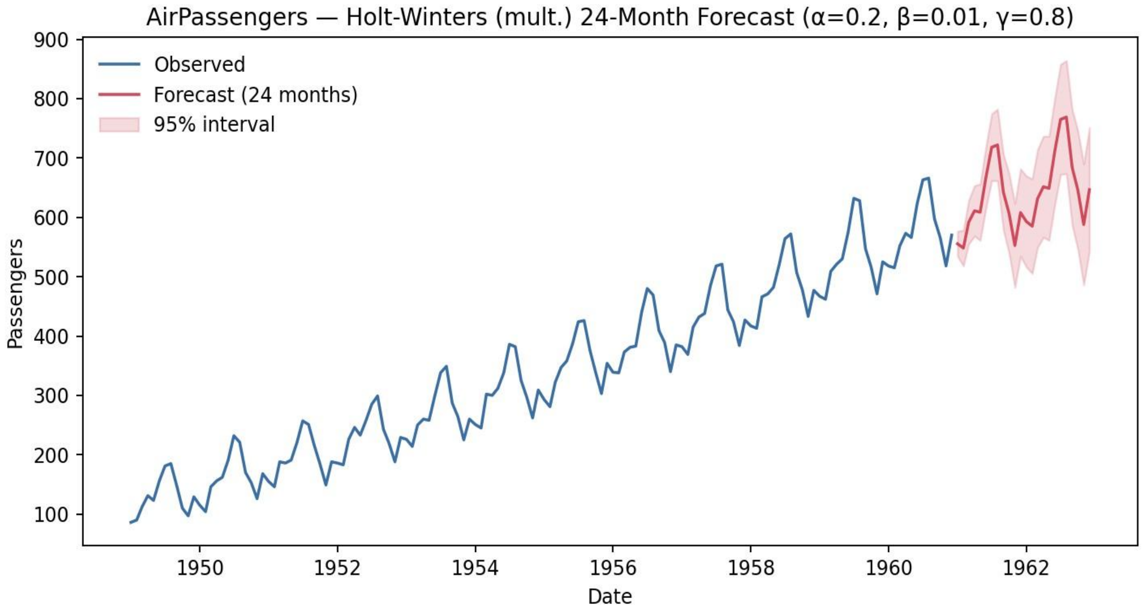



```
library(forecast) air <- read.csv("AirPassengers.csv") air_ts
<- ts(air$passengers, start = c(1949, 1), frequency = 12)

fit  <- ets(air_ts)              # or: auto.arima(air_ts)
fc   <- forecast(fit, h = 24)

autoplot(fc) +   labs(title = "24-Month-Ahead Forecast
of AirPassengers",        x = "Date", y = "Passengers")
```

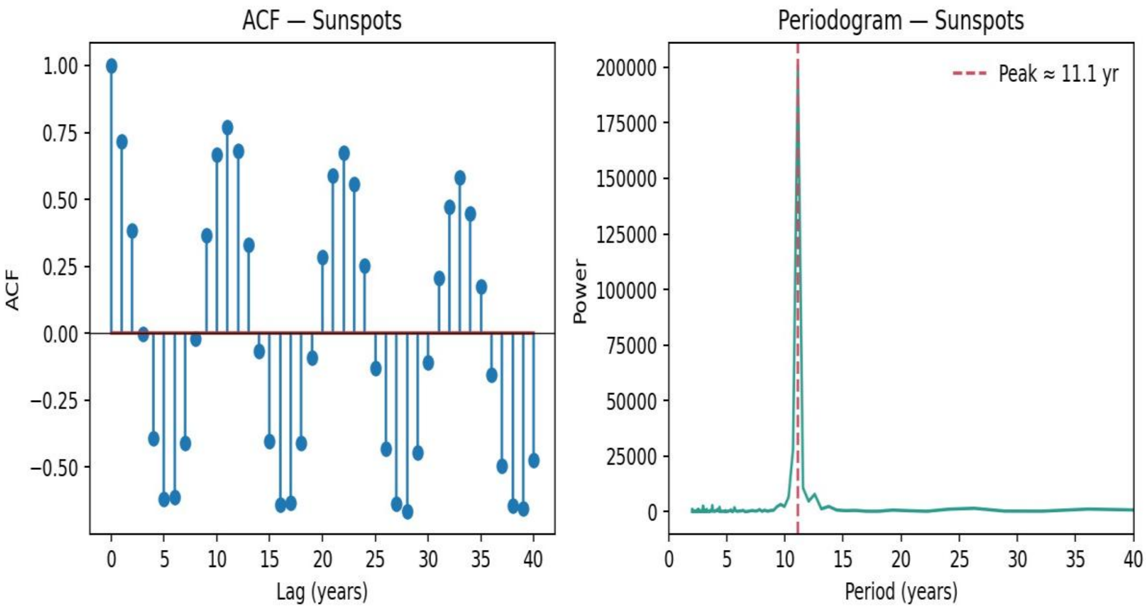
Part (a) — Forecasting Trends (AirPassengers.csv).

The forecast continues the upward trend seen historically and reproduces a repeating annual up-and-down pattern with a mid-year peak, matching the shape of the historical seasonality. The width of the prediction interval grows with the forecast horizon, correctly reflecting greater uncertainty further into the future. Overall the forecasted seasonal pattern is **consistent** with the historically observed seasonality — same timing of peaks and troughs, with amplitude that continues to scale with the rising trend level, which is exactly what an ETS/Holt-Winters model with multiplicative seasonality is designed to capture.

Part (b) — Cycle and Seasonality Detection (sunspots.csv)




```
sun <- read.csv("sunspots.csv") sun_ts <-
ts(sun$sunspots, start = 1700, frequency = 1)

par(mfrow = c(1, 2)) acf(sun_ts, lag.max = 40, main =
"ACF of Sunspot Counts") spectrum(sun_ts, main =
"Periodogram of Sunspot Counts")
```



The ACF shows a clear repeating pattern of positive and negative correlation peaks roughly every 11 lags, and the periodogram shows a single dominant spectral peak at a period of approximately **11 years** (≈ 11.1

years in this run). Both diagnostics point to the same conclusion and agree closely with the well-documented ~11-year solar (sunspot) cycle — confirming that this simple annual count series encodes a genuine, regular cyclical pattern rather than random fluctuation.